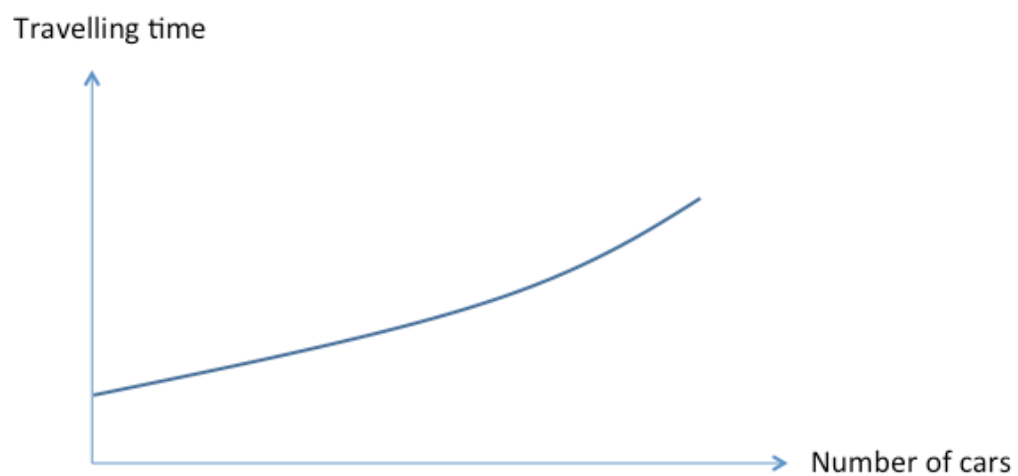


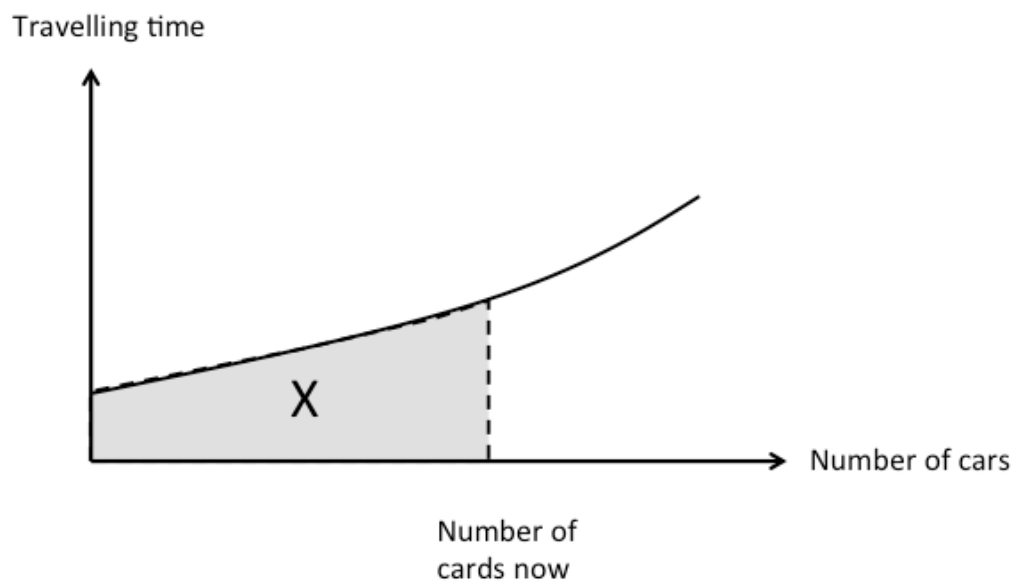
Question: Is it possible that people always switch to a better road but never reach a Nash Equilibrium?

Answer: No, they always reach Nash equilibrium in the traffic game!

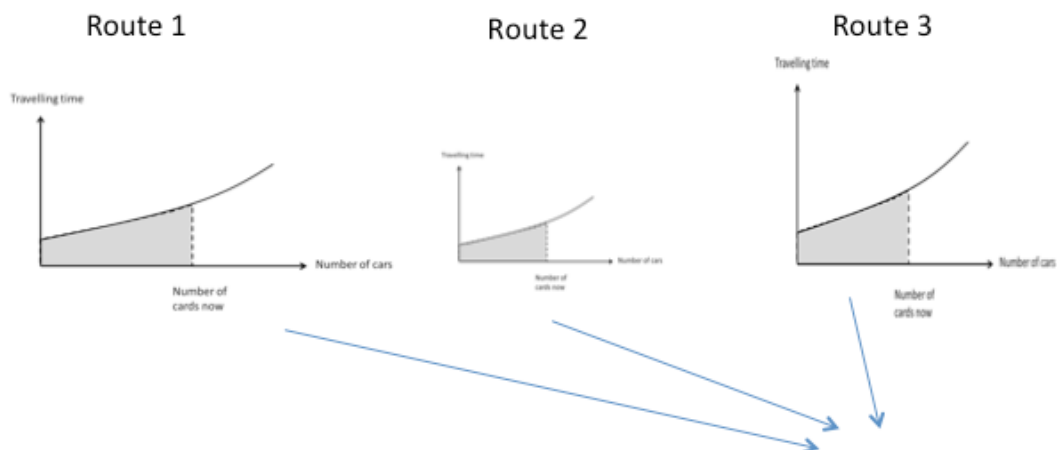
Let me show you why....



In each route, the travelling time depends on the number of cars.



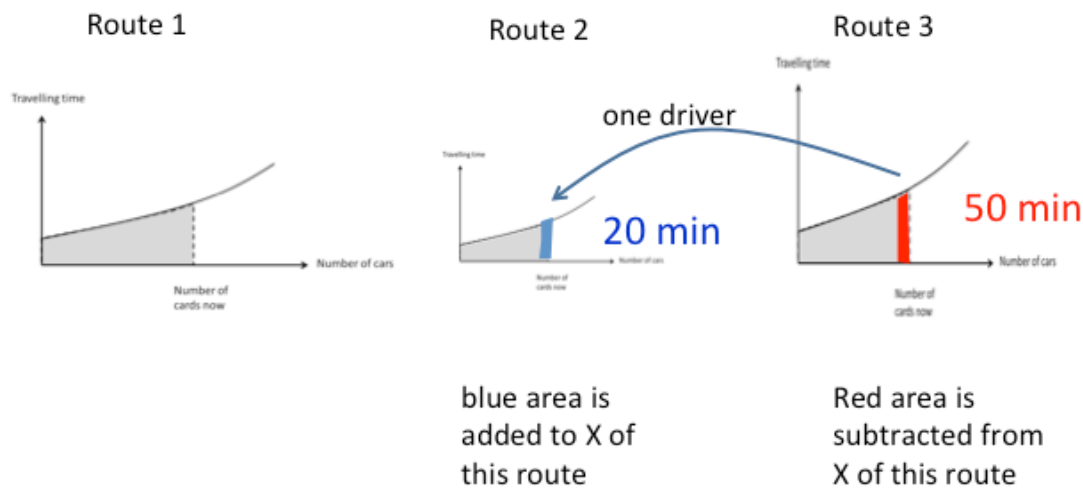
Now calculate the area X, for each route.



... and sum up "X"s for all routes and call it **P**.

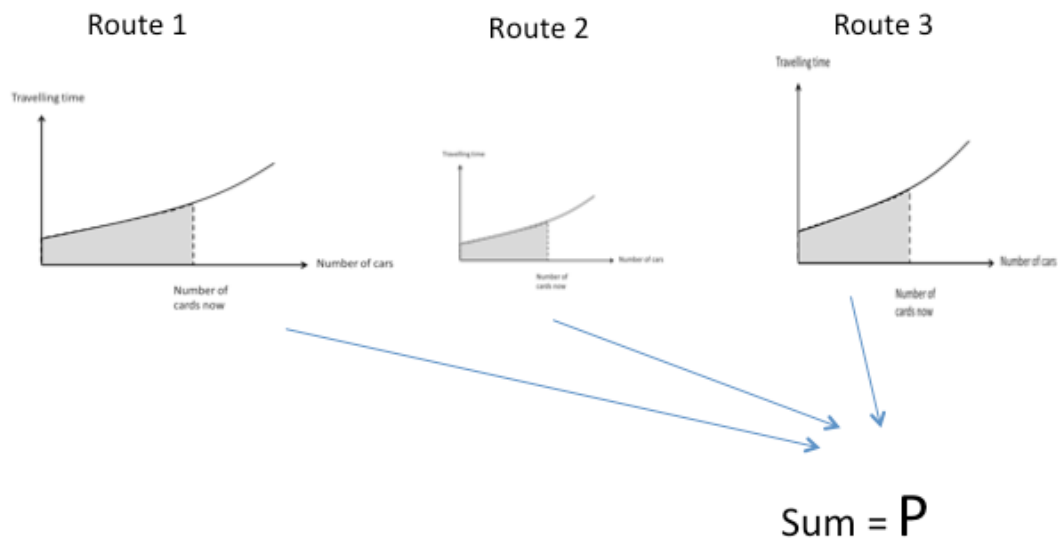
If there are three routes, P is the sum of those three gray areas.

Now suppose a driver switches from route 3 to route 2 and his/her traveling time is reduced from 50 min to 20 min. Then the gray areas of routes 2 and 3 change as follows:



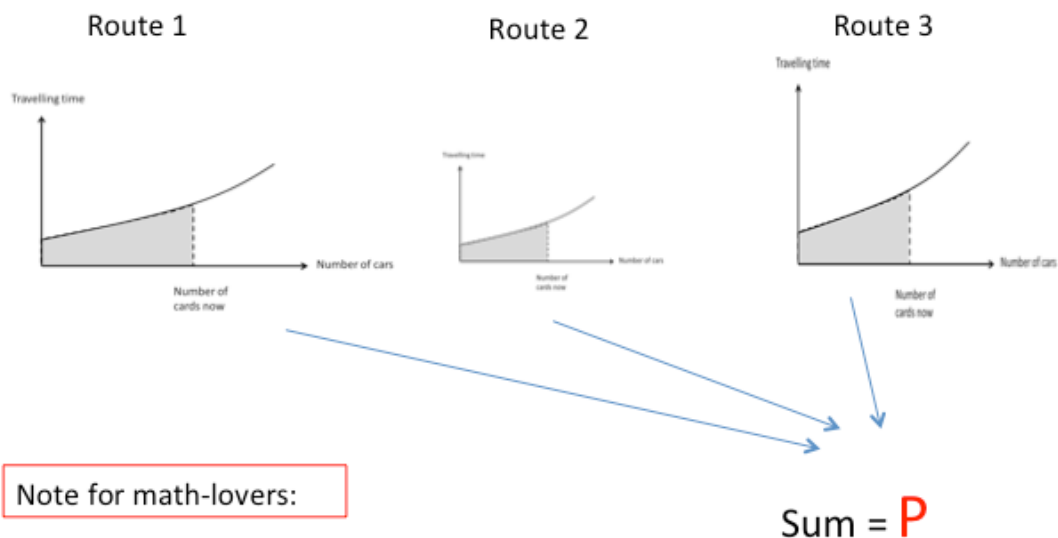
Hence P , the sum of gray areas, changes by $-50 + 20 = -30$.

P is reduced when a driver switches to a better route !



Therefore, P decreases as drivers switch to better routes, and eventually we reach a situation where there is no room to P make smaller (i.e., there is no better route for any one). That is the Nash equilibrium!

The upshot is that the Nash equilibrium IS ALWAYS reached by the adjustment of routes of drivers.



Note for math-lovers:

P is called the “**potential**” of the traffic game.

Believe it or not, when we switch to a better route, we are minimizing the potential of the traffic game!

